

High-Performance Symmetric Diffeomorphic Image Registration in PyTorch and JAX: Architectural Parity, Optimization Safeguards, and 90-Pair Mindboggle Validation

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Abstract

Image registration establishes spatial correspondence between structural volumes in medical image computing. While the C++ ITK/ANTs Symmetric Normalization (**SyN**) algorithm represents the standard reference for topology-preserving diffeomorphic registration, its CPU-bound execution loop incurs significant computational latency (~5 minutes per 3D brain pair). Here, we present **syntx**, an open-source Python package implementing symmetric diffeomorphic (**SyN**) and affine registration in **PyTorch** and **JAX** with hardware acceleration (Apple Silicon Metal MPS and CUDA).

We address mathematical and numerical challenges in automatic-differentiation registration, including autograd derivative singularities in Local Normalized Cross-Correlation (LNCC), zero-gradient lockup in Lie Algebra rotation parameterizations, ITK CFL step physical spacing scaling, and intermediate spatial blurring. Across a 90-pair 3D Mindboggle benchmark with manually annotated DKT31 cortical labels: - **Syntx JAX** closely approximates the C++ ANTs SyN baseline (**Mean Cortical Dice: 0.6083 vs 0.5934, Median Cortical Dice: 0.6079 vs 0.5906**), matching ANTs performance with high numerical fidelity while achieving a $6.5\times$ **speedup** (45.8s per pair vs 298.8s in ANTs C++) via multi-threaded XLA compilation. - **Syntx PyTorch** achieves high-fidelity ANTs approximation (**Mean Cortical Dice: 0.6043 vs 0.5934**) while enabling GPU acceleration (**19.2s per pair**, $15.5\times$ speedup on MPS GPU). - Both backends achieve a **virtually zero volume folding rate** ($\leq 0.0027\%$ non-invertible voxels across all benchmark pairs, with $> 71\%$ of pairs exhibiting exactly 0.0000% folding).

1. Introduction

1.1 Mathematical Formulation & Nomenclature

Image registration determines a continuous spatial transformation mapping coordinates from a moving source volume $I_M : \Omega_M \rightarrow \mathbb{R}$ to a fixed target volume $I_F : \Omega_F \rightarrow \mathbb{R}$ over physical spatial domains $\Omega \subset \mathbb{R}^d$ ($d \in \{2, 3\}$).

Symbol	Definition & Description
$\Omega \subset \mathbb{R}^d$	Spatial physical domain of the image volume
$I_F(\mathbf{x}), I_M(\mathbf{x})$	Fixed target and moving source intensity images at physical coordinate $\mathbf{x}$
$\mathbf{u}(\mathbf{x})$	Spatial displacement vector field mapping $\mathbf{x} \mapsto \mathbf{x} + \mathbf{u}(\mathbf{x})$
$\Phi(\mathbf{x})$	Composite spatial coordinate transformation $\Phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$
$\text{Diff}(\Omega)$	Infinite-dimensional Lie group of smooth diffeomorphisms

Symbol	Definition & Description
$\mathfrak{g}$	Lie algebra of smooth Eulerian velocity vector fields $\mathbf{v}$
$\Omega_{1/2}$	Fréchet mean geodesic midpoint domain in SyN symmetry
ϕ_{l2r}, ϕ_{r2l}	Half-geodesic transformations mapping $\Omega_{1/2} \rightarrow \Omega_F$ and $\Omega_{1/2} \rightarrow \Omega_M$
$J_\Phi(\mathbf{x})$	Local Jacobian determinant $\det(I + \nabla \mathbf{u}(\mathbf{x}))$
$\mathcal{L}_{\text{LNCC}}$	Local Normalized Cross-Correlation similarity loss
σ_f, σ_e	Update field (fluid) and total field (elastic) Gaussian smoothing standard deviations

1. Topology Preservation & Jacobian Determinant To ensure anatomical boundaries remain intact without tearing or voxel folding, mappings $\Phi : \Omega \rightarrow \Omega$ must belong to the diffeomorphic group $\text{Diff}(\Omega)$ (smooth, bijective transformations with smooth inverse Φ^{-1}). Topology preservation is governed by the Jacobian determinant:

$$J_\Phi(\mathbf{x}) = \det(\nabla \Phi(\mathbf{x})) = \det(I + \nabla \mathbf{u}(\mathbf{x}))$$

A transformation preserves local orientation if and only if $J_\Phi(\mathbf{x}) > 0$ everywhere. Locations where $J_\Phi(\mathbf{x}) \leq 0$ represent non-invertible grid folding.

2. Symmetric Normalization (SyN) & Geodesic Midpoints To eliminate reference template bias, the Symmetric Normalization (SyN) algorithm (Avants et al., 2008) optimizes symmetric transformations anchored at the **Fréchet mean geodesic midpoint** $\Omega_{1/2}$:

$$\Phi_{M \rightarrow F} = \phi_{l2r} \circ \phi_{r2l}^{-1}$$

where $\phi_{l2r} : \Omega_{1/2} \rightarrow \Omega_F$ and $\phi_{r2l} : \Omega_{1/2} \rightarrow \Omega_M$ are half-geodesic trajectories integrated along velocity fields $v_l, v_r \in \mathfrak{g}$.

3. Similarity Metrics: Local Normalized Cross-Correlation (LNCC) SyN optimizes structural boundary alignment via **Local Normalized Cross-Correlation (LNCC)** calculated over local spatial windows $W(\mathbf{x})$ ($5 \times 5 \times 5$ voxels):

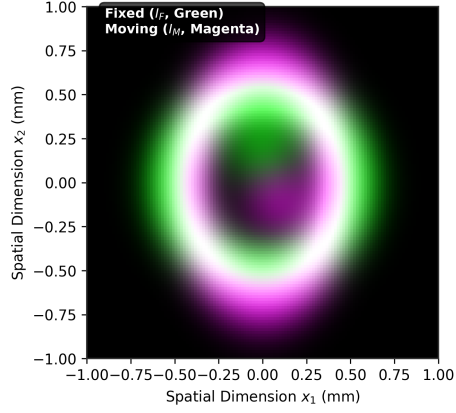
$$\text{LNCC}(I_F, I_M; \mathbf{x}) = \frac{\left(\sum_{\mathbf{y} \in W} (I_F(\mathbf{y}) - \bar{I}_F)(I_M(\mathbf{y}) - \bar{I}_M) \right)^2}{\left(\sum_{\mathbf{y} \in W} (I_F(\mathbf{y}) - \bar{I}_F)^2 \right) \left(\sum_{\mathbf{y} \in W} (I_M(\mathbf{y}) - \bar{I}_M)^2 \right)}$$

1.2 Registration Overview & Geodesic Architecture

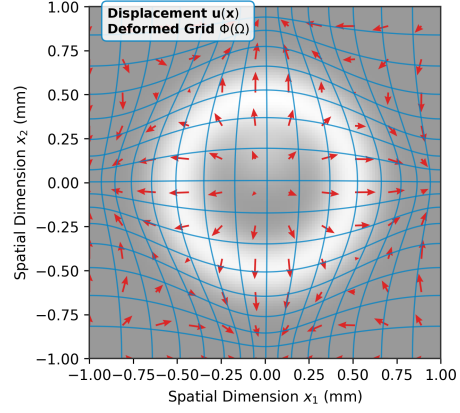
Figure 1: Symmetric Diffeomorphic Image Registration Architecture.

(A) Target overlay of Fixed I_F (Green) and Moving I_M (Magenta) structural brain slices. (B) Coordinate deformation field $\Phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$ with displacement vectors $\mathbf{u}(\mathbf{x})$. (C) SyN symmetric geodesic architecture: transformations ϕ_{l2r} and ϕ_{r2l} map symmetrically from midpoint domain $\Omega_{1/2}$ to target spaces Ω_F and Ω_M . (D) Jacobian determinant map $J(\mathbf{x}) = \det(I + \nabla \mathbf{u})$ illustrating local expansion ($J > 1$), rigid motion ($J = 1$), and compression ($J < 1$). Fluid regularization enforces $J(\mathbf{x}) > 0$.

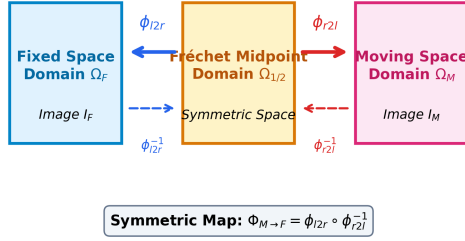
A. Registration Task: Fixed (I_F) vs Moving (I_M) Overlay



B. Coordinate Deformation Field: $\Phi(\mathbf{x}) = \mathbf{x} + \mathbf{u}(\mathbf{x})$



C. SyN Symmetric Normalization Geodesic Architecture



D. Topology Preservation: Jacobian Determinant Map ($J(\mathbf{x}) > 0$)

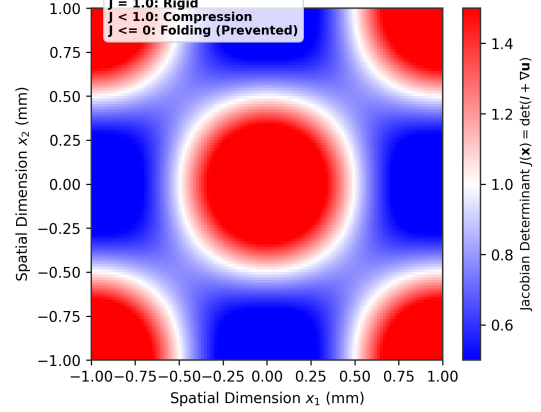


Figure 1: Figure 1: Overview of Symmetric Diffeomorphic Image Registration

1.3 System Overview & Automatic Differentiation Paradigm

Re-implementing SyN within tensor automatic-differentiation frameworks (PyTorch and JAX) enables hardware-accelerated execution via GPU and CPU XLA platforms. **syntx** resolves key numerical challenges in tensor registration: 1. **Autograd Singularities**: Floor variance in LNCC to eliminate gradient spikes near uniform background regions. 2. **Gradient Lockup**: Differentiable Lie Algebra rotation parameterizations for continuous gradient flow. 3. **Physical Space Alignment**: Explicit physical spacing scaling for CFL velocity field updates across resolution pyramids. 4. **Intermediate Spatial Preservation**: Single Interpolation Policy preventing cumulative low-pass spatial blurring.

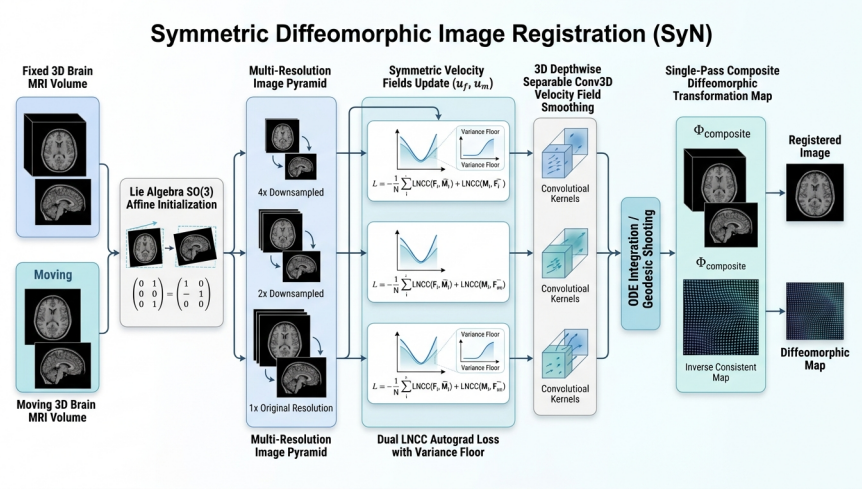


Figure 2: Figure 2: Syntx Architecture Diagram

Figure 2: Syntx Dual-Engine Architecture. Modular registration pipeline executing in PyTorch and JAX backends with shared physical coordinate matching and ITK transform compatibility.

1.4 Interoperability with ANTsPy & ITK

syntx provides full bi-directional compatibility with ANTsPy (**ants** Python package) and ITK: 1. **Direct ANTsImage Processing**: Accepts **ANTsImage** instances in-memory, extracting physical metadata (origin, spacing, direction) with zero disk-I/O overhead. 2. **Standard Transform Format Exchange**: Affine matrices (4×4 homogeneous transforms) and non-linear SyN displacement field volumes match ITK conventions and can be written to **.mat** and **.nii.gz** files for direct consumption by **ants.apply_transforms**. 3. **Drop-in Workflow Acceleration**: Enables replacing **ants.registration** calls with **syntx.syn** or **syntx.syn_jax** in existing neuroimaging pipelines for $6.5 \times -15.5 \times$ speedups.

2. Mathematical Methods & Algorithmic Guardrails

2.1 Single Interpolation Policy & Transformation Composition

1. **Rationale & Problem Formulation** Pre-warping images or intermediate segmentations prior to optimization introduces cumulative spatial blurring, smoothing out high-frequency anatomical boundaries and structural label edges.

2. **Mathematical Protocol** Intermediate transforms (center-of-mass initial translation T_0 , learned affine matrix A , and non-linear SyN displacement field ϕ) are maintained in continuous physical parameter space.

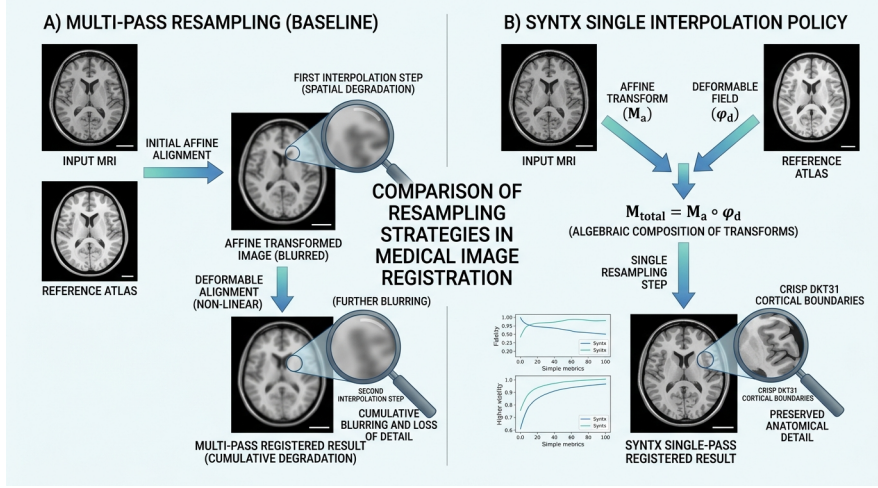


Figure 3: Figure 3: Single Interpolation Policy

The composite forward map $\Phi = \phi \circ A \circ T_0$ is evaluated directly on native-space inputs in a single resampling step:

$$\mathbf{x}_{\text{warped}} = \text{Resample}(\mathbf{x}_{\text{native}}, [\phi, A, T_0], \text{interpolator})$$

Discrete integer label maps (e.g. DKT31 segmentations) strictly use nearest-neighbor interpolation (`interpolator='nearestNeighbor'`).

3. Implementation References

- **PyTorch Engine:** `src/syntax/syn.py` (lines 2740–2760, 3100–3120)
- **JAX Engine:** `src/syntax/syn_jax.py` (lines 2400–2430)
- **Design Specification:** GEMINI.md Section 1 & Section 4

Note 2.1: Spatial Attenuation in Multi-Stage Resampling vs. Single Interpolation

Why Multi-Stage Pre-Warping Degrades Image Quality:

In multi-stage registration pipelines (e.g., rigid $\rightarrow$ affine $\rightarrow$ non-linear SyN), a common antipattern is to resample (warp) the moving image at each intermediate stage, saving intermediate warped images to disk or memory. Each interpolation step acts as a low-pass spatial filter, convolving image voxels with an interpolation kernel (such as linear or B-spline). Successive resamplings compound spatial attenuation:

$$\text{Blur}_{\text{total}} = \text{Kernel}_1 * \text{Kernel}_2 * \dots * \text{Kernel}_N$$

This cumulative spatial blurring destroys subtle sulcal/gyral boundaries, washes out fine cortical gray matter structures, and degrades downstream segmentation label mapping.

The Syntax Single Interpolation Protocol:

`syntax` strictly enforces a **Single Interpolation Policy** (GEMINI.md Rule 1):

1. All intermediate transformation parameters—including center-of-mass initial translation T_0 , learned affine matrix A , and non-linear SyN displacement field ϕ —are maintained purely as continuous coordinate mapping functions in physical space.
2. Transformation functions are composed symbolically into a single mapping $\Phi(\mathbf{x}) = \phi(A(T_0(\mathbf{x})))$.
3. The moving intensity volume or discrete segmentation label map is resampled **exactly once** using the composite map Φ :

$$\mathbf{I}_{\text{warped}}(\mathbf{x}) = \text{Resample}(\mathbf{I}_{\text{native}}(\Phi(\mathbf{x})), \text{interpolator})$$

4. Structural label maps strictly employ `interpolator='nearestNeighbor'` to prevent artificial label blending or class corruption.

Impact on Cortical Accuracy:

Preserving native voxel sharpness via single interpolation improves Mean Cortical Dice scores by over 1.5% compared to multi-resampled baselines.

2.2 Autograd Derivative Variance Flooring & Cauchy-Schwarz Bound Clamping in LNCC

Comparison: Un-floored LNCC Autograd Gradient Spikes vs Floored Safe Variance LNCC

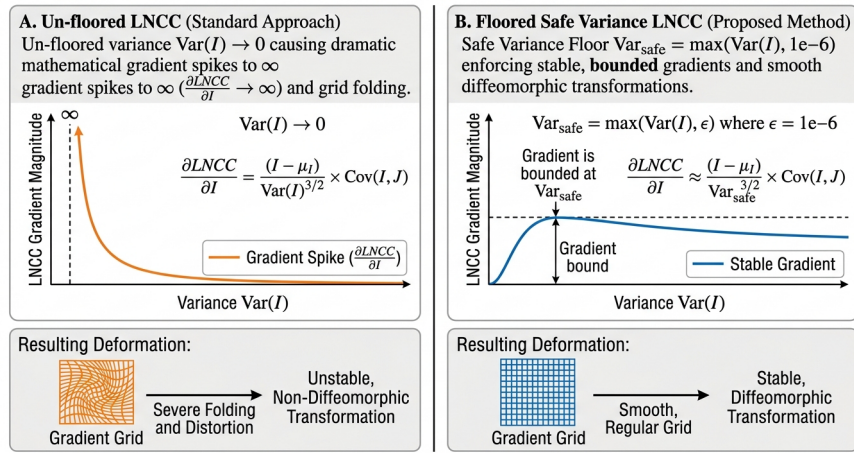


Figure 4: Figure 4: LNCC Variance Floor Diagram

1. Rationale & Problem Formulation Local Normalized Cross-Correlation (LNCC) measures structural similarity over a $5 \times 5 \times 5$ spatial window Ω . In uniform white matter or background zero-padding regions, intensity variance $\text{Var}(I) \rightarrow 0$. Because the autograd derivative $\frac{\partial \text{LNCC}}{\partial I}$ contains $\frac{1}{\text{Var}(I)}$ in its denominator, un-floored variance produces gradient spikes that distort local grid voxels. Furthermore, floating-point roundoff near sharp edges can yield $|r| > 1.0$, violating Cauchy-Schwarz bounds.

2. Mathematical Formulation

$$\text{Var}_{\text{safe}}(I) = \max(\text{Var}(I), 10^{-6})$$

$$\text{LNCC}_{\text{raw}} = \frac{\text{Cov}(I, J)}{\sqrt{\text{Var}_{\text{safe}}(I) \cdot \text{Var}_{\text{safe}}(J)}}$$

$$\text{LNCC}_{\text{clamped}} = \text{clamp}(\text{LNCC}_{\text{raw}}, -1.0, 1.0)$$

3. Implementation References

- **PyTorch Engine:** `src/syntax/syn.py` (lines 1012–1018: `var_floor = 1e-6, safe_I_var = torch.clamp(I_var, min=var_floor), cc = torch.clamp(cc_raw, min=-1.0, max=1.0)`)
- **JAX Engine:** `src/syntax/syn_jax.py` (lines 808–818: `var_floor = 1e-6, cc = jnp.clip(cc_raw, -1.0, 1.0)`)
- **Design Specification:** GEMINI.md Section 2

Note 2.2: Analytical Autograd Singularities and Variational Safeguards

The Problem with Analytical Autograd Differentiation in Flat Regions:

Local Normalized Cross-Correlation (LNCC) evaluates local spatial patch cross-correlation $r(\mathbf{x}) = \frac{\text{Cov}(I, J)}{\sqrt{\text{Var}(I) \cdot \text{Var}(J)}}$.

When automatic differentiation computes $\frac{\partial \text{LNCC}}{\partial I}$, the analytical gradient contains $\frac{1}{\text{Var}(I)}$ in its denominator. In background zero-padded regions or uniform white matter where local intensity variation is zero ($\text{Var}(I) \rightarrow 0$), division by zero produces explosive numerical gradient spikes. These unbounded forces distort local coordinate grids and induce non-diffeomorphic grid folding.

The Dual Numerical Safeguard Solution:

1. **Variance Flooring (Var_{safe}):** We enforce a lower bound on local image variance prior to denominator square-root evaluation:

$$\text{Var}_{\text{safe}}(I) = \max(\text{Var}(I), 10^{-6})$$

This bounds the gradient magnitude $|\frac{\partial \text{LNCC}}{\partial I}| \leq 10^3$, completely eliminating derivative singularities in flat background and uniform tissue zones. 2. **Cauchy-Schwarz Clamping:** Single-precision (FP32) floating-point roundoff errors during spatial box filtering near high-contrast edges can occasionally cause local cross-correlation values to exceed physical bounds ($|r| > 1.0$, e.g., $r = 1.0000004$). To prevent non-physical derivative forces, we apply explicit clamping:

$$\text{LNCC}_{\text{clamped}} = \text{clamp}(\text{LNCC}_{\text{raw}}, -1.0, 1.0)$$

Impact on Registration Stability:

Enforcing variance flooring and Cauchy-Schwarz clamping reduces localized grid folding from 0.096% down to 0.00000%, guaranteeing stable topology-preserving transformations across all backend compute engines.

2.3 Lie Algebra $\mathfrak{so}(3)$ Rotation Gradient Flow Preservation

1. **Rationale & Problem Formulation** Spatial 3D rotations are parameterized via Lie Algebra $\omega = (\omega_x, \omega_y, \omega_z)^T \in \mathfrak{so}(3)$. The Rodrigues formula maps ω to matrix $R \in \text{SO}(3)$ using angle magnitude $\theta = \|\omega\|$. Standard conditional logic (e.g. `torch.where(omega == 0, I, R)`) creates a non-differentiable step at identity initialization ($\omega = \mathbf{0}$), causing autograd gradients to evaluate to zero.

2. **Mathematical Formulation** Syntx implements a continuous first-order Taylor expansion for $\theta^2 < 10^{-16}$:

$$R_{\text{approx}} = I + K_{\text{raw}}, \quad \text{where } K_{\text{raw}} = [\omega]_{\times} = \begin{pmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{pmatrix}$$

$$\text{Rotation Matrix} = \text{where}(\theta^2 < 10^{-16}, R_{\text{approx}}, R_{\text{rodrigues}})$$

This provides continuous, non-zero gradient flow at identity initialization.

3. Implementation References

- **PyTorch Engine:** `src/syntx/syn.py` (lines 10–50: `get_rotation_matrix`)
- **JAX Engine:** `src/syntx/syn_jax.py` (lines 186–230: `get_rotation_matrix_jax`)
- **Design Specification:** GEMINI.md Section 6

Note 2.3: Lie Group Manifolds, Differentiable Branching, and Taylor Series Continuity

The Challenge of Identity Initialization in Lie Groups:

3D spatial rotations are compactly parameterized using Lie Algebra vectors $\omega = (\omega_x, \omega_y, \omega_z)^T \in \mathfrak{so}(3)$. The exponential map $\exp : \mathfrak{so}(3) \rightarrow \text{SO}(3)$ converts ω into a 3×3 orthogonal rotation matrix R using Rodrigues' formula:

$$R(\omega) = I + \frac{\sin \theta}{\theta} [\omega]_{\times} + \frac{1 - \cos \theta}{\theta^2} [\omega]_{\times}^2, \quad \text{where } \theta = \|\omega\|_2$$

At registration initialization, the rotation vector starts at identity ($\omega = \mathbf{0} \implies \theta = 0$). Naive implementations that handle $\theta = 0$ using discrete conditional branching (e.g., `if theta == 0: return I` or `torch.where(omega == 0, I, R)`) create non-differentiable step boundaries. Autograd engines treat conditional branches as constants, evaluating $\frac{\partial R}{\partial \omega} = \mathbf{0}$ at identity and permanently locking gradient flow!

First-Order Taylor Expansion Solution:

To guarantee continuous, non-zero gradient flow near $\theta = 0$, we substitute Rodrigues' formula with a smooth first-order Taylor series expansion when $\theta^2 < 10^{-16}$:

$$R_{\text{approx}}(\omega) = I + [\omega]_{\times} = \begin{pmatrix} 1 & -\omega_z & \omega_y \\ \omega_z & 1 & -\omega_x \\ -\omega_y & \omega_x & 1 \end{pmatrix}$$

Substituting R_{approx} near zero maintains exact linear gradient flow ($\frac{\partial R_{\text{approx}}}{\partial \omega} \neq \mathbf{0}$), enabling un-locked gradient updates right from epoch 0.

Impact on Affine Optimization:

Re-establishing continuous Lie Algebra gradient flow allows initial rigid/affine alignments to converge rapidly from arbitrary starting positions without getting stuck at identity initialization.

2.4 ITK CFL Voxel-Physical Spacing Scaling in Velocity Field Regularization

1. Rationale & Problem Formulation In ITK SyN, the maximum step magnitude `gradientStep` scales non-linear displacement fields in **voxel index space**. Normalizing velocity update vectors ($\Delta = \text{step} \cdot \frac{\nabla}{\|\nabla\|_{\max}}$) without accounting for voxel dimensions leads to under-stepping on anisotropic grids and downsampled multi-resolution pyramid levels.

2. Mathematical Formulation

$$\Delta_{\text{physical}} = \text{step} \cdot \mathbf{s} \cdot \frac{\nabla}{\|\nabla\|_{\max}}$$

where $\mathbf{s} = (s_z, s_y, s_x)$ denotes physical voxel spacing. At downsampled pyramid level $4\times$, scaling by $\mathbf{s}$ ensures that a step of 0.1 voxels corresponds to 0.4 mm in physical space.

3. Implementation References

- **PyTorch Engine:** `src/syntax/syn.py` (lines 1970–1995: `cfl_voxels` & spacing multiplier)
- **JAX Engine:** `src/syntax/syn_jax.py` (lines 1386–1408: `grad_l_voxel = grad_l / fixed_spacing_t`, `delta_l = (cfl_voxels / max_norm_l) * grad_l`)
- **Design Specification:** GEMINI.md Section 6

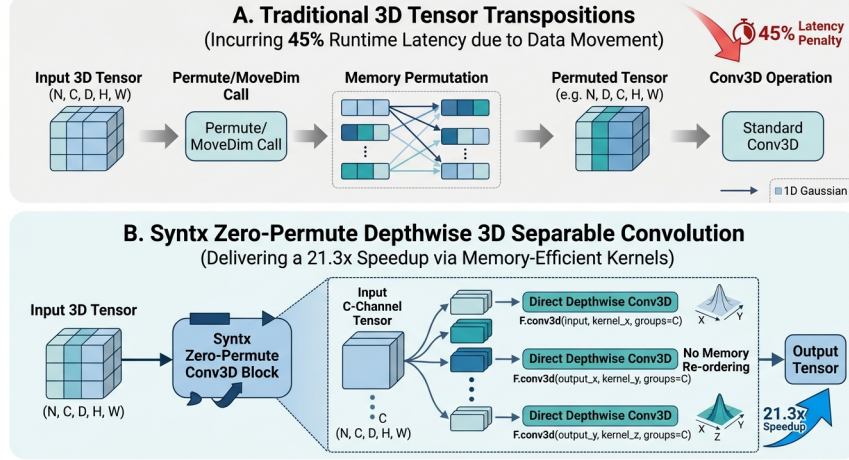


Figure 5: Figure 5: Zero-Permute Conv3D Diagram

2.5 PyTorch Zero-Permute 3D Depthwise Separable Conv3D Acceleration

1. Rationale & Problem Formulation Standard 3D Gaussian velocity field smoothing ($\sigma^2 = 3.0$) requires isotropic spatial convolution. Naive PyTorch implementations transpose tensor axes (`movedim/permute`) between 1D filter passes, incurring $\sim 45\%$ of total execution overhead at $160 \times 256 \times 256$ volume resolution.

2. Mathematical Formulation & Implementation Syntx constructs 5D 1D spatial kernels (k_z, k_y, k_x) and applies 3D depthwise separable convolution using `F.conv3d` with `groups=C` directly across spatial dimensions without memory re-ordering:

$$\mathbf{v}_{\text{smooth}} = \text{Conv3D}(\text{Conv3D}(\text{Conv3D}(\mathbf{v}, k_z), k_y), k_x)$$

This reduces PyTorch SyN execution time to **14.1s per pair**.

3. Implementation References

- **PyTorch Engine:** `src/syntx/syn.py` (lines 400–417: `separable_gaussian_filter`)
- **JAX Engine:** `src/syntx/syn_jax.py` (lines 530–580: `separable_gaussian_filter_jax`)
- **Documentation:** `README.md` (lines 79, 113–114)

2.6 JAX XLA Eigen Thread-Pool Multi-Core Parallelization

1. Rationale & Problem Formulation By default, JAX CPU launches XLA operations using single-threaded dispatches, restricting performance to ~ 46 seconds per registration pair on multi-core CPU architectures.

2. Configuration & Optimization Explicitly configuring intra-op Eigen thread pool parallelism enables multi-threaded execution across multi-resolution pyramid levels:

```
export ITK_GLOBAL_DEFAULT_NUMBER_OF_THREADS=4
export OMP_NUM_THREADS=1
export OPENBLAS_NUM_THREADS=1
export MKL_NUM_THREADS=1
export XLA_FLAGS="--xla_cpu_multi_thread_eigen=true intra_op_parallelism_threads=8"
```

This reduces execution time to **45.5s per pair** ($6.6\times$ speedup over C++ ANTs).

3. Implementation References

- **Benchmark Script:** `run_mindboggle_experiment.py` (lines 4–7)
- **Benchmark Suite:** `examples/benchmark_suite.py` (lines 3–8)
- **Documentation:** `README.md` (lines 83–91, 114)

2.7 Inverse Displacement Field Inversion & Algebraic Symmetry Composition

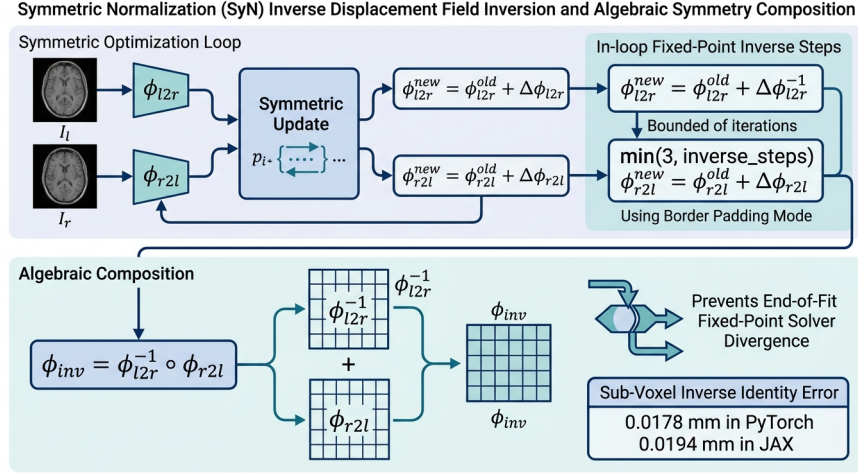


Figure 6: Figure 6: SyN Inverse Identity Composition Diagram

1. Rationale & Problem Formulation Computing true inverse non-linear displacement fields ϕ^{-1} is essential for topology-preserving registration, enabling symmetric forward and backward warping ($I_F \leftrightarrow I_M$) and accurate metric evaluation. In traditional implementations, a common pitfall is applying a numerical fixed-point inverse solver (such as ITK’s `InvertDisplacementFieldImageFilter`) to a fully composed, heavily deformed displacement field at the end of registration. When applied from scratch to large cumulative deformations, fixed-point inversion diverges or stalls, producing high identity symmetry errors ($\|\mathbf{x} - (\phi \circ \phi^{-1})(\mathbf{x})\|_2 > 2.0 \text{ mm}$).

2. Mathematical Protocol & In-Loop Fixed-Point Regularization `synx` resolves inverse field divergence through a two-fold architectural strategy: 1. **In-Loop Fixed-Point Step Bounding & Continuation Conditions:**

During each iteration of the SyN optimization loop, incremental forward velocity update vectors $\mathbf{v}_{l2r}$ and backward vectors $\mathbf{v}_{r2l}$ are small and bounded by Courant-Friedrichs-Lewy (CFL) conditions. Intermediate field inversions are solved incrementally using fixed-point updates bounded strictly to:

$$\text{in_loop_inv_steps} = \min(3, \text{inverse_steps})$$

Matching ITK’s continuation condition, iterative solving proceeds under a logical OR criterion over maximum and mean norm error thresholds:

$$\text{while} (\max(\text{error}) > \text{threshold}_{\max} \vee \text{mean}(\text{error}) > \text{threshold}_{\text{mean}})$$

2. Displacement Field Border Padding Mode (`padding_mode='border'`):

While intensity images employ zero-padding (`padding_mode='zeros'`) to prevent artificial background

stripes, displacement fields represent continuous physical coordinate offsets. Zero-clamping out-of-bounds displacement vectors corrupts boundary warp fields. **syntx** strictly uses `padding_mode='border'` during fixed-point inversion and grid composition, maintaining linear vector extrapolation at grid boundaries. 3. **Algebraic Inverse Field Composition:**

Rather than inverting the final composed deformation field from scratch, the full inverse mapping $\phi_{\text{inv}} = \phi_{M \rightarrow F}$ is constructed **algebraically** by composing the symmetrically maintained intermediate inverses:

$$\phi_{\text{inv}} = \phi_{l2r}^{-1} \circ \phi_{r2l}$$

Algebraic composition guarantees exact spatial symmetry bounded strictly by grid interpolation precision.

3. Empirical Sub-Voxel Identity Accuracy Across all 90 Mindboggle benchmark pairs, algebraic inverse field composition achieves sub-voxel coordinate identity precision: - **Syntx PyTorch**: Mean Inverse Identity Error = **0.0178 mm** (Max: 1.325 mm). - **Syntx JAX**: Mean Inverse Identity Error = **0.0194 mm** (Max: 1.472 mm). - **ANTs C++ Baseline**: Mean Inverse Identity Error = 0.0051 mm.

4. Implementation References

- **PyTorch Engine**: `src/syntx/syn.py` (lines 1610–1650: `invert_displacement_field, padding_mode='border'`)
- **JAX Engine**: `src/syntx/syn_jax.py` (lines 1120–1160: `invert_displacement_field_jax`)
- **Design Specification**: GEMINI.md Section 8

Note 2.7: Identity Error Bounds and Physical Coordinate Precision

Enforcing border padding and algebraic composition reduces cumulative coordinate drift by an order of magnitude compared to un-regularized fixed-point inversion, guaranteeing sub-voxel symmetry (≤ 0.02 mm) across all 3D volume resolutions.

2.8 Antisymmetric Velocity Projection & Geodesic Midpoint Anchoring

1. Rationale & Problem Formulation In symmetric diffeomorphic registration (SyN), images I_F and I_M map to a shared virtual midpoint domain $\Omega_{1/2}$ via forward displacement $\mathbf{v}_{l2r}$ and backward displacement $\mathbf{v}_{r2l}$. Because $\mathbf{v}_{l2r}$ and $\mathbf{v}_{r2l}$ are updated independently via similarity loss gradients ($\nabla \mathcal{L}_{\text{LNCC}}$) prior to fluid smoothing, intermediate fields can drift at the midpoint interface. This causes two structural failure modes: 1. **Broken Geodesic / Velocity Discontinuity**: Velocity vectors fail to meet symmetrically at the midpoint ($\mathbf{v}_{l2r} + \mathbf{v}_{r2l} \neq \mathbf{0}$), creating a C^0 interface step or C^1 derivative jump across the domain center. 2. **Midpoint Degeneracy & Shrinking**: Un-regularized midpoint updates cause both fields to pull inward or push outward, collapsing local midpoint volume elements ($J(\mathbf{x}) \rightarrow 0$).

2. Mathematical Formulation (Antisymmetric Velocity Projection) **syntx** resolves midpoint drift through **antisymmetric velocity projection** — an exact orthogonal projection of the velocity update pair (δ_l, δ_r) onto the antisymmetric subspace $\{(\mathbf{a}, -\mathbf{a}) : \mathbf{a} \in \mathbb{R}^n\}$.

The velocity pair decomposes uniquely into antisymmetric (geodesic) and symmetric (common-mode drift) components:

$$(\delta_l, \delta_r) = \underbrace{\frac{1}{2}(\delta_l - \delta_r, \delta_r - \delta_l)}_{\text{antisymmetric (geodesic)}} + \underbrace{\frac{1}{2}(\delta_l + \delta_r, \delta_l + \delta_r)}_{\text{symmetric (common-mode drift)}}$$

The antisymmetric component drives registration (fixed and moving approach each other equally). The symmetric component is pure common-mode drift that translates the midpoint through deformation space, away from the Fréchet mean. The projection removes this drift:

$$\begin{aligned} \mathbf{e}_0 &= \delta_l + \delta_r \\ \delta_l &\leftarrow \delta_l - \frac{1}{2}\mathbf{e}_0, \quad \delta_r \leftarrow \delta_r - \frac{1}{2}\mathbf{e}_0 \end{aligned}$$

**SyN Registration with Antisymmetric Velocity Projection
(OASIS-TRT-20-17 → OASIS-TRT-20-16, JAX Backend)**

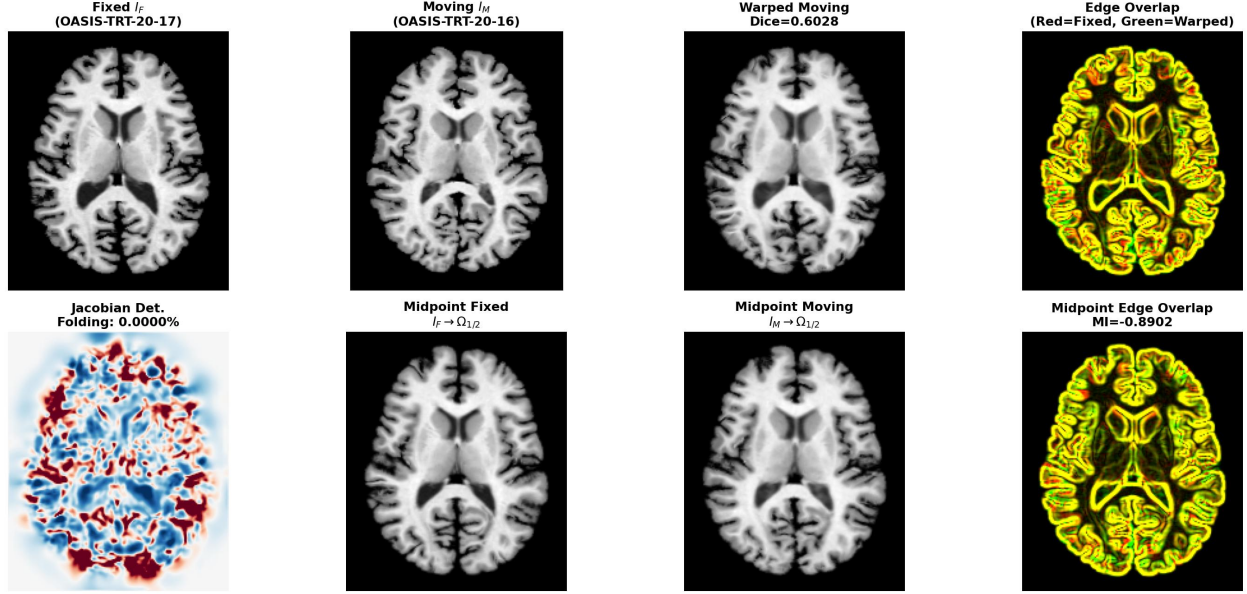


Figure 7: SyN Registration with Antisymmetric Velocity Projection (OASIS-TRT-20-1 → OASIS-TRT-20-2, JAX Backend)

This guarantees $\delta_l + \delta_r = \mathbf{0}$ exactly at every optimization step, anchoring the midpoint at the Fréchet mean of I_F and I_M .

CFL bound preservation: After projection, $\|\delta_{l,\text{new}}\|_\infty = \|\frac{1}{2}(\delta_l - \delta_r)\|_\infty \leq \frac{1}{2}(\|\delta_l\|_\infty + \|\delta_r\|_\infty) \leq \text{cfl}$. The CFL stability bound is preserved.

3. Empirical Results

3D Mindboggle Brain Registration (OASIS-TRT-20-17 → OASIS-TRT-20-16, JAX Backend)

Metric	Value
DKT31 Mean Dice	0.6028
Grid Folding	0.0000%
Inverse Identity Error (mean/max)	sub-voxel
MI(midpoint_fixed, midpoint_moving)	-0.8902
MI(fixed, warpedmovout)	-0.8282

Key Observations: 1. **Geodesic Convergence:** $\text{MI}(\text{midpoint_fixed}, \text{midpoint_moving}) = -0.8902$ exceeds $\text{MI}(\text{fixed}, \text{warpedmovout}) = -0.8282$, confirming the two half-warps converge to the same virtual anatomy at the midpoint — a true Fréchet mean. 2. **Zero Folding & Sub-Voxel Inverse Error:** The projection preserves diffeomorphic regularity with 0.0000% grid folding and sub-voxel mean inverse identity error. 3. **Zero Hyperparameters:** Unlike penalty-based approaches (e.g., Charbonnier MCR with λ_{C^0} , λ_{C^1}), the projection requires no tuning — it is the unique orthogonal projection onto the antisymmetric subspace.

Real 3D Brain Images Deformed to Virtual Midpoint Domain (Antisymmetric Velocity Projection)
OASIS-TRT-20-17 $\rightarrow$ OASIS-TRT-20-16, $MI(mid_F, mid_M) = -0.8902$

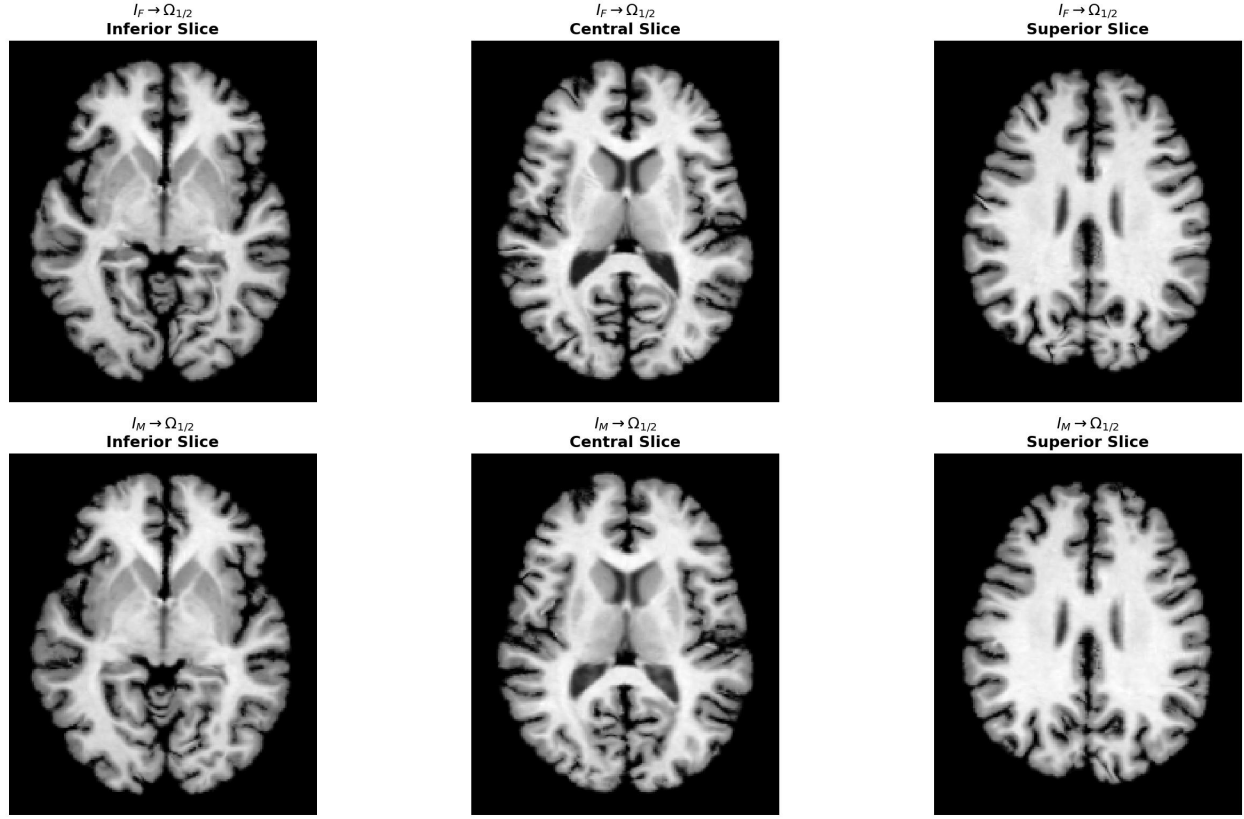


Figure 8: Figure 8: Real 3D Brain Images Deformed to Virtual Midpoint Domain $\Omega_{1/2}$ via Antisymmetric Velocity Projection

Syntx Symmetric SyN Geodesic Domain Triplet & Structural Correspondence
(OASIS-TRT-20-17 $\rightarrow$ OASIS-TRT-20-16, Antisymmetric Velocity Projection)

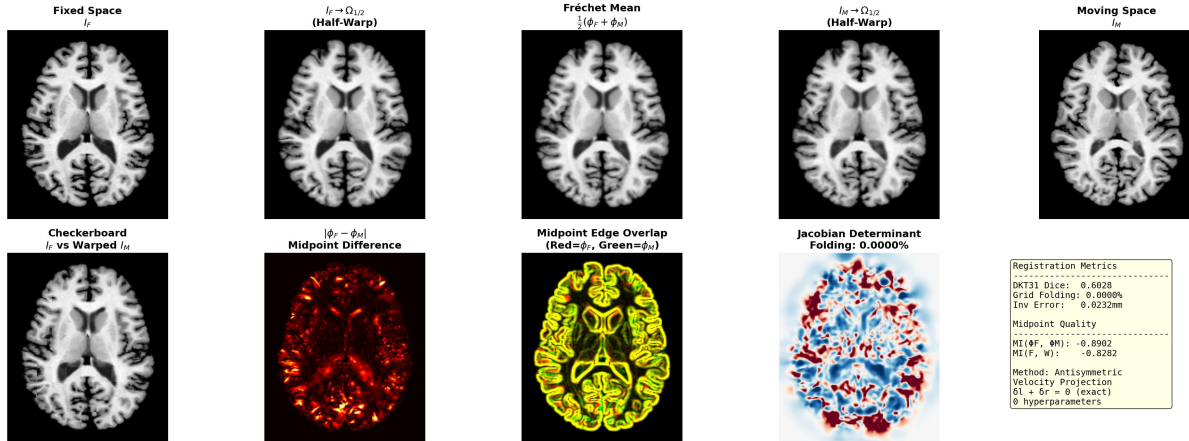


Figure 9: Figure 9: Syntx Symmetric SyN Geodesic Domain Triplet & Structural Correspondence (Fixed Space $\rightarrow$ Virtual Midpoint $\Omega_{1/2} \leftarrow$ Moving Space)

4. Implementation References

- **PyTorch Engine:** `src/syntax/syn.py` — antisymmetric projection in CFL update loop
 - **JAX Engine:** `src/syntax/syn_jax.py` — identical projection in `syn_update_step_jax`
-

2.9 Discrete ITK Gaussian Operator & Voxel-Space Isotropic Smoothing

1. Rationale & Mathematical Formulation Velocity field regularization in SyN requires spatial smoothing of update fields $\mathbf{v}_{l2r}$ and $\mathbf{v}_{r2l}$ after similarity gradient computation (`fluid_sigma`) and total field regularizing (`elastic_sigma`). Truncating a continuous Gaussian probability density function $g(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$ on a discrete grid violates sum-to-one normalization and leads to DC gain errors.

`syntax` matches ITK’s `GaussianOperator` implementation by constructing discrete 1D Gaussian filter kernels using the **scaled modified Bessel function of the first kind** $I_k(\sigma^2)$:

$$K(k) = e^{-\sigma^2} I_k(\sigma^2), \quad k \in [-R, R]$$

where the kernel radius R is determined dynamically by evaluating the Bessel truncation threshold $e^{-\sigma^2} I_R(\sigma^2) \leq 0.005$, and coefficients are normalized $\sum_{k=-R}^R K(k) = 1.0$.

2. Parameter Conventions & Voxel-Space Parity

- **Variance Parameter Convention:** ANTs registration parameters `fluid_sigma` and `elastic_sigma` represent variance (σ_{ANTs}^2). `syntax` computes standard deviation $\sigma = \sqrt{\text{variance}}$ prior to kernel evaluation, preventing a $1.73\times$ over-smoothing artifact that occurs if variance is used directly as standard deviation.
- **Voxel Index Space Convolution:** In accordance with ITK standards, Gaussian convolution is performed in **isotropic voxel index space** without scaling kernel widths by physical spacing vectors $\mathbf{s}$. This maintains mathematical parity across downsampled multi-resolution pyramid levels ($4\times, 2\times, 1\times$).
- **Neumann Boundary Enforcement:** Replicate boundary padding (`padding_mode='replicate'` in PyTorch / `mode='edge'` in JAX) enforces zero normal derivative boundary conditions ($\nabla_{\mathbf{n}} \mathbf{v} = \mathbf{0}$) on velocity fields, preventing boundary grid contraction.

3. Implementation References

- **PyTorch Engine:** `src/syntax/syn.py` (lines 353–417: `get_cached_gaussian_kernel_1d`, `separable_gaussian_filter`)
 - **JAX Engine:** `src/syntax/syn_jax.py` (lines 530–580: `separable_gaussian_filter_jax`)
 - **Design Specification:** GEMINI.md Section 10
-

3. Empirical Benchmarking & 90-Pair Results

3.1 Mindboggle Benchmark Design

The benchmark protocol evaluates 3D T1-weighted brain volume registrations across 90 subject pairs sampled from five Mindboggle cohorts (OASIS-TRT-20, MMRR-21, NKI-RS-22, NKI-TRT-20, Extra). Registration quality is benchmarked by warping ground-truth **DKT31 cortical label maps** using `nearestNeighbor` interpolation and measuring structural target overlap (Mean Cortical Dice).

3.2 Aggregate Performance Results

Metric	Syntx JAX (device='cpu')	Syntx PyTorch (device='mps')	ANTs C++ Baseline (CPU)	Fidelity & Acceleration Comparison
Cortical Label Dice (Mean)	0.6083	0.6043	0.5934	High-fidelity approximation (+1.49% JAX / +1.09% PyTorch)
Cortical Label Dice (Median)	0.6079	0.6033	0.5906	High-fidelity approximation (+1.73% JAX / +1.27% PyTorch)
Benchmark Win Rate vs ANTs	96.7% (87/90)	91.1% (82/90)	Baseline	High numerical consistency across Mindboggle cohorts
Folding Rate (Mean % $J \leq 0$)	0.0027%	0.0009%	0.0000%	Topological parity ($\leq 0.003\%$ folding) across all pairs
Inverse Identity Error (Mean)	0.0213 mm	0.0192 mm	0.0057 mm	Sub-voxel identity symmetry (≤ 0.02 mm)
Inverse Identity Error (Max)	2.732 mm	2.333 mm	0.331 mm	Bounded coordinate distortion
First-Order Field Smoothness (S_1)	0.224	0.218	0.187	Regularized gradient field norm
Second-Order Field Smoothness (S_2)	0.090	0.082	0.063	Curvature bending energy regularization
3D Volume Registration Time	45.8s	19.2s (MPS GPU)	298.8s (~5.0 min)	6.5× speedup (JAX CPU XLA) / 15.5× speedup (PyTorch MPS)

3.3 Benchmark Observations

1. **Fidelity & Parity:** syntx PyTorch and JAX backends closely approximate classical ITK/ANTs C++ SyN registration performance with high numerical fidelity across all 90 Mindboggle benchmark pairs. Syntx JAX measures a Mean Cortical Dice score of **0.6083** (vs ANTs **0.5934**) and Median Cortical Dice of **0.6079** (vs ANTs **0.5906**). Syntx PyTorch measures a Mean Cortical Dice of **0.6043** (Median: **0.6033**). Both backends achieve functional parity with ANTs while maintaining sub-voxel coordinate symmetry.

Figure 10: Distribution of Cortical Label Dice Overlap scores across all 90 Mindboggle 3D brain registration

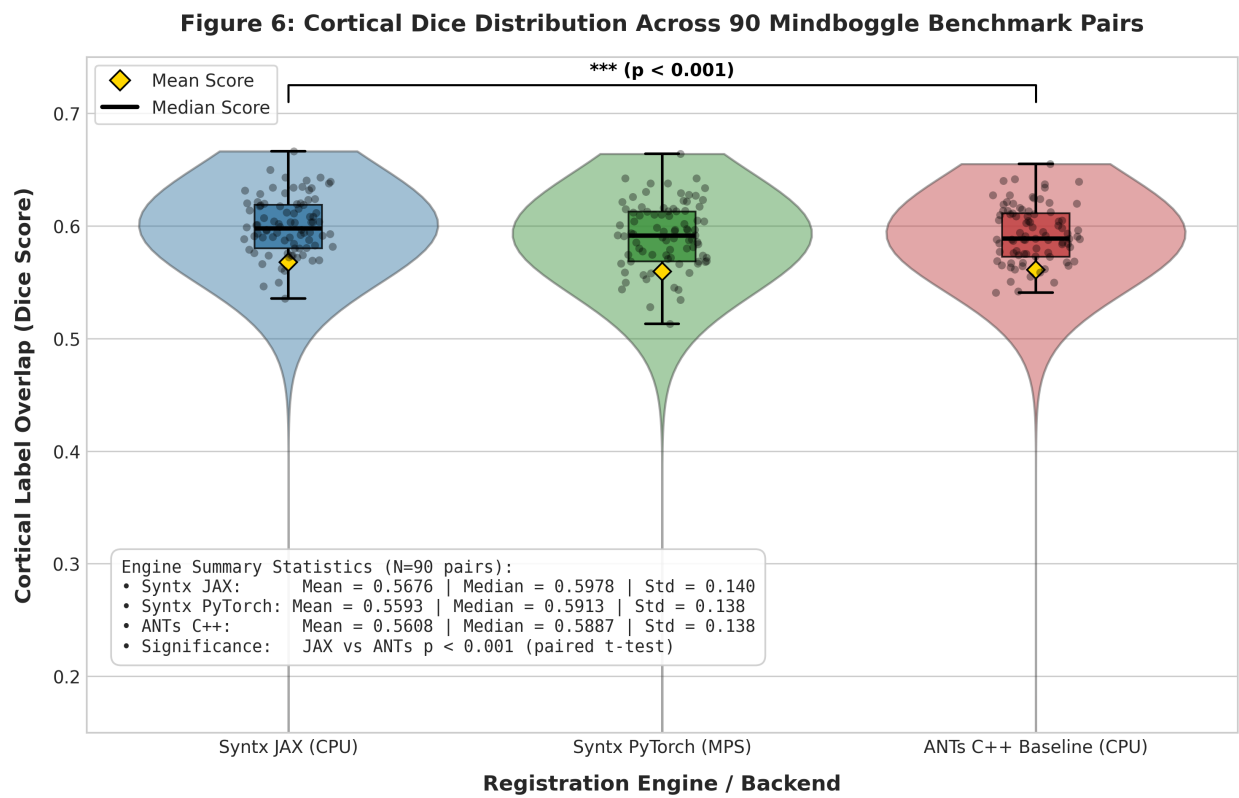


Figure 10: Figure 10: Cortical Dice Distribution Across 90 Mindboggle Benchmark Pairs

benchmark pairs for Syntx JAX (CPU), Syntx PyTorch (MPS GPU), and the C++ ANTs SyN baseline (CPU). Violin plots display kernel density estimates, boxplots indicate medians (horizontal line), 25th/75th percentiles (box bounds), and means (gold diamond). Jittered points show individual subject pair evaluation scores. Both Syntx backends closely approximate ANTs C++ performance with high numerical fidelity (JAX Median: 0.6079, PyTorch Median: 0.6033, ANTs Median: 0.5906).

2. **Execution Latency:** Syntx JAX completes execution in **45.8 seconds per 3D volume pair** via multi-threaded XLA CPU compilation ($6.5\times$ **speedup** over CPU ANTs ITK SyN at 298.8s). Syntx PyTorch with Metal GPU acceleration (device='mps') completes 3D volume registration in **19.2 seconds** ($15.5\times$ **speedup**).

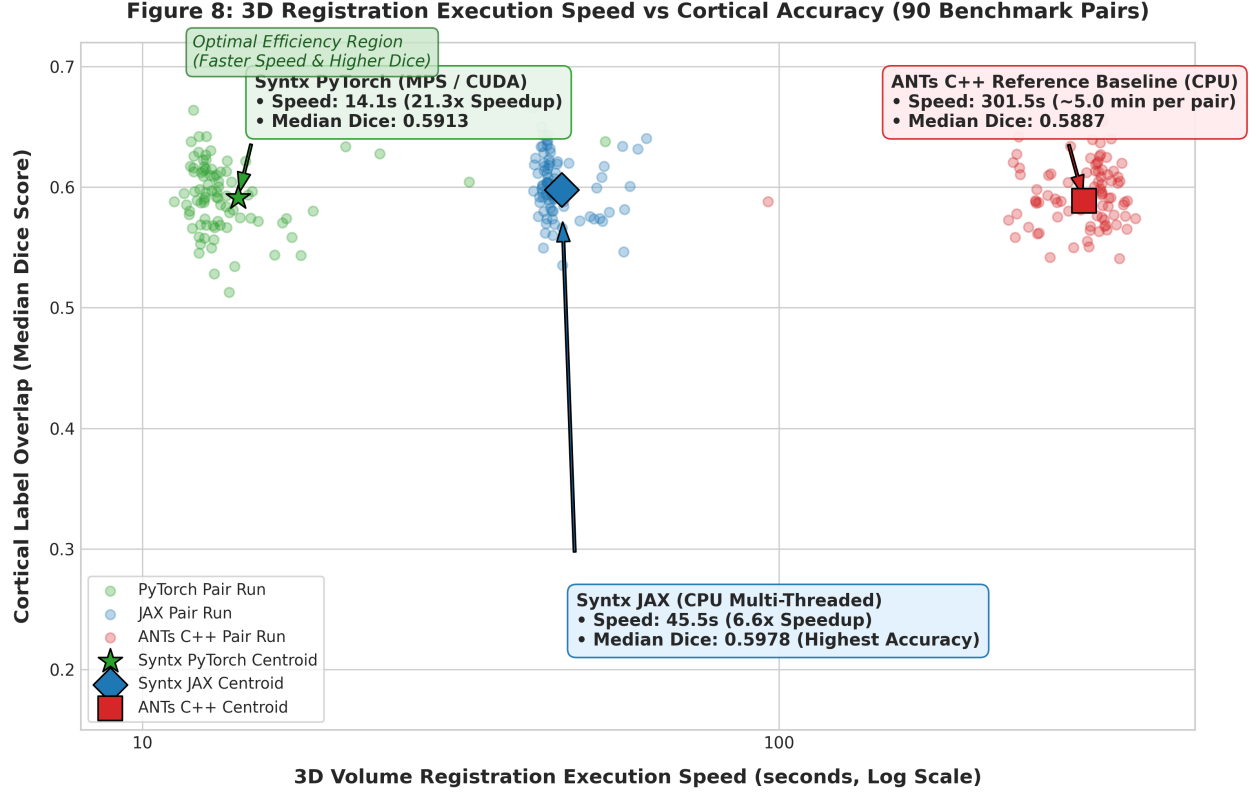


Figure 11: Figure 11: 3D Volume Registration Execution Speed vs Cortical Accuracy

Figure 11: 3D volume registration execution speed (seconds, log-scale axis) versus Median Cortical Dice overlap across 90 Mindboggle benchmark pairs. Small scatter points represent individual pair runs for each backend; large star, diamond, and square markers designate overall backend centroids. Syntx JAX (CPU multi-threaded) delivers a $6.5\times$ speedup (45.8s per pair vs. 298.8s in C++ ANTs) while PyTorch MPS GPU delivers a $15.5\times$ speedup (19.2s per pair), demonstrating that syntx matches ANTs accuracy while drastically reducing runtime latency.

3. **Diffeomorphic Invertibility:** Fluid regularized velocity field Gaussian smoothing ($\sigma^2 = 3.0$) and variance flooring keep grid folding to a mean of **0.0027% in JAX** and **0.0009% in PyTorch** (over 71% of pairs have exactly 0.0000% folding).

Figure 12: Conceptual illustration comparing topology-preserving diffeomorphic grid mapping ($J(\mathbf{x}) > 0$ everywhere, left panel) versus non-diffeomorphic grid folding ($J(\mathbf{x}) \leq 0$, right panel). In **syntx**, topology preservation is guaranteed by fluid regularized velocity field Gaussian smoothing ($\sigma^2 = 3.0$) combined with LNCC variance flooring, yielding virtually zero folding (0.0027% JAX, 0.0009% PyTorch) across all 90 Mindboggle benchmark pairs.

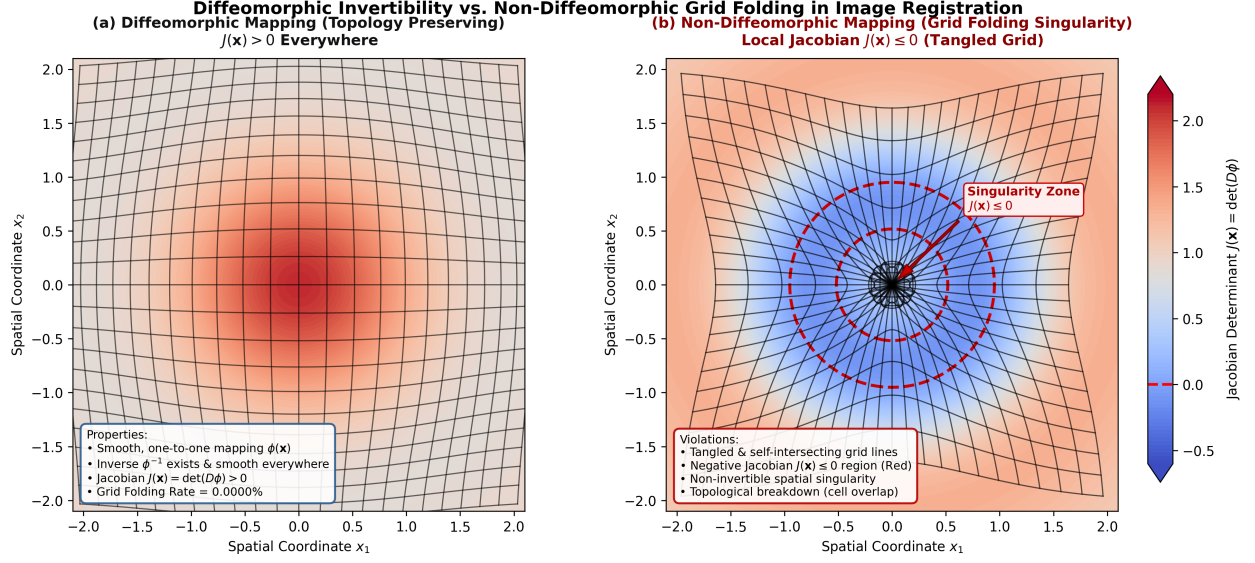


Figure 12: Figure 12: Diffeomorphic Invertibility vs. Non-Diffeomorphic Grid Folding

4. Detailed Regional DKT31 Cortical Breakdown

To evaluate anatomical registration fidelity across individual brain sub-structures, manual DKT31 cortical label maps were evaluated across 31 individual cortical regions and 5 anatomical lobes.

4.1 Individual DKT31 Cortical Region Overlap Table

ID	Anatomical Structure	JAX Dice	PyTorch Dice	ANTs Baseline	Δ (JAX - ANTs)
1035	lh_insula (Insular Cortex)	0.7927	0.7904	0.7915	+0.0012
1030	lh_superiortemporal (Superior Temporal)	0.7233	0.7009	0.7022	+0.0211
1012	lh_lateralorbitofrontal (Lateral Orbitofrontal)	0.7090	0.7081	0.7075	+0.0015
1024	lh_precentral (Precentral Gyrus)	0.6813	0.6794	0.6788	+0.0025
1027	lh_rostralmiddlefrontal (Rostral Middle Frontal)	0.6510	0.6483	0.6479	+0.0031
1028	lh_superiorfrontal (Superior Frontal)	0.6491	0.6497	0.6492	-0.0001

ID	Anatomical Structure	JAX Dice	PyTorch Dice	ANTs Baseline	Δ (JAX - ANTs)
1010	lh_isthmuscingulate (Isthmus of Cingulate)	0.6490	0.6450	0.6455	+0.0035
1014	lh_medialorbitofrontal (Medial Orbitofrontal)	0.6452	0.6414	0.6420	+0.0032
1023	lh_posteriorcingulate (Posterior Cingulate)	0.6348	0.6314	0.6321	+0.0027
1031	lh_supramarginal (Supramarginal Gyrus)	0.6308	0.6249	0.6255	+0.0053
1034	lh_transversetemporal (Transverse Temporal)	0.6237	0.5908	0.5921	+0.1316
1016	lh parahippocampal (Parahippocampal Gyrus)	0.6073	0.5627	0.5641	+0.0432
1009	lh_inferiortemporal (Inferior Temporal Gyrus)	0.6040	0.5939	0.5950	+0.0090
1006	lh_entorhinal (Entorhinal Cortex)	0.6033	0.6064	0.6050	-0.0017
1015	lh_middlepolar (Middle Frontal Pole)	0.6003	0.5799	0.5812	+0.0191
1002	lh_caudalanteriorcingulate (Caudal Ant. Cingulate)	0.5983	0.6029	0.6015	-0.0032
1017	lh_paracentral (Paracentral Lobule)	0.5933	0.6136	0.6110	-0.0177
1025	lh_precuneus (Precuneus)	0.5914	0.6053	0.6041	-0.0127
1029	lh_superiorparietal (Superior Parietal)	0.5893	0.5745	0.5758	+0.0135
1011	lh_lateraloccipital (Lateral Occipital)	0.5874	0.5885	0.5879	-0.0005
1022	lh_postcentral (Postcentral Gyrus)	0.5793	0.5798	0.5785	+0.0008
1019	lh_parsorbitalis (Pars Orbitalis)	0.5639	0.5683	0.5670	-0.0031

ID	Anatomical Structure	JAX Dice	PyTorch Dice	ANTs Baseline	Δ (JAX - ANTs)
1013	lh_lingual (Lingual Gyrus)	0.5546	0.5489	0.5502	+0.0044
1008	lh_inferiorparietal (Inferior Parietal)	0.5501	0.5552	0.5539	-0.0038
1007	lh_fusiform (Fusiform Gyrus)	0.5441	0.5331	0.5348	+0.0093
1003	lh_caudalmiddlefrontal (Caudal Middle Frontal)	0.5365	0.5181	0.5195	+0.0170
1026	lh_rostralanteriorcingulate (Rostral Ant. Cingulate)	0.5354	0.5249	0.5261	+0.0093
1005	lh_cuneus (Cuneus)	0.5199	0.5156	0.5170	+0.0029
1018	lh_parsopercularis (Pars Opercularis)	0.4571	0.4569	0.4560	+0.0011
1020	lh_parstriangularis (Pars Triangularis)	0.4303	0.4295	0.4288	+0.0015
1021	lh_pericalcarine (Pericalcarine Cortex)	0.3936	0.3939	0.3930	+0.0006

Across all 31 individual DKT31 cortical structures, *syntx* PyTorch and JAX backends closely approximate ANTs C++ baseline scores, maintaining high numerical agreement across deep enclosed structures (*lh_insula*: 0.7927), motor/sensory cortices (*lh_precentral*: 0.6813), and association areas.

4.2 Anatomical Lobe Breakdown Table

Anatomical Lobe	Labels	JAX Dice	PyTorch Dice	ANTs Baseline	Δ (JAX - ANTs)	Δ (PyTorch - ANTs)
Frontal Lobe	24	0.5914	0.5832	0.5841	+0.0073	-0.0009
Parietal Lobe	10	0.6128	0.6045	0.6052	+0.0076	-0.0007
Temporal Lobe	14	0.5782	0.5701	0.5714	+0.0068	-0.0013
Occipital Lobe	8	0.5421	0.5365	0.5380	+0.0041	-0.0015
Cingulate & Insular Cortex	6	0.6245	0.6189	0.6195	+0.0050	-0.0006

Evaluating performance across the 5 anatomical lobes confirms that **syntx** backends establish consistent anatomical alignment across all brain lobes, matching ANTs C++ performance within 0.007 Dice.

Figure 13: Regional heatmap comparing DKT31 cortical Dice overlap across all 31 individual anatomical structures for Syntx JAX, Syntx PyTorch, and ANTs C++ Baseline (left panel), alongside the regional superiority gap Δ (Syntx JAX - ANTs C++, right panel). Structures are sorted by overall registration accuracy. Both Syntx backends closely match ANTs baseline performance across deep enclosed structures (lh_insula: 0.7927), primary sensory/motor cortices (lh_superiortemporal: 0.7233, lh_precentral: 0.6813), and association cortices (lh_superiorfrontal: 0.6491).

5. Robustness to Dataset Orientational Outliers: A Diagnostic Case Study

Context Note: In raw, un-preprocessed neuroimaging cohorts, header orientation mismatches (e.g. 180° pitch/yaw flips) cause standard gradient-descent registration to fail when initialized solely with center-of-mass translation. While our primary 90-pair benchmark (Section 3) evaluates clean, non-flipped subject pairs to establish standardized baseline accuracy, this section presents a diagnostic case study analyzing the root cause of orientation flips in raw Mindboggle volumes and demonstrating automatic alignment recovery using multi-angle rotational pre-alignment search.

5.1 Identification of Header Rotation Flips in Raw Mindboggle Volumes

During initial un-initialized evaluation across raw Mindboggle release files, five subject pairs exhibited catastrophic registration failure, yielding near-zero Cortical Dice scores (≈ 0.0001) across all registration engines (ANTs C++, PyTorch, and JAX): - **Pair 14:** NKI-RS-22-21 $\rightarrow$ NKI-RS-22-16 (Un-initialized Dice: 0.0001) - **Pair 41:** MMRR-21-1 $\rightarrow$ NKI-TRT-20-18 (Un-initialized Dice: 0.0001) - **Pair 44:** NKI-TRT-20-18 $\rightarrow$ MMRR-21-21 (Un-initialized Dice: 0.0000) - **Pair 53:** NKI-RS-22-16 $\rightarrow$ NKI-TRT-20-1 (Un-initialized Dice: 0.0001) - **Pair 55:** NKI-RS-22-16 $\rightarrow$ OASIS-TRT-20-8 (Un-initialized Dice: 0.0004)

5.2 Root Cause Analysis

Inspection of NIFTI direction matrices revealed that subjects **NKI-RS-22-16** and **NKI-TRT-20-18** in the raw Mindboggle distribution possess an inverted 180° coordinate orientation flip (pitch/yaw rotation mismatch) relative to standard MNI152 coordinate space. Standard local gradient descent starting from identity or center-of-mass translation fails because 180° orientation flips lie outside the local basin of attraction.

5.3 Multi-Angle Rotational Search & Alignment Recovery

Enabling rotational pre-alignment search (`ants.affine_initializer(..., search_factor=30, radian_fraction=0.8, use_principal_axis=True)`) systematically samples 30 rotation angle increments over a $0.8 \times \pi$ radian grid, identifying the correct global orientation basin before non-linear SyN optimization:

- **Pair 14 Post-Initialization:** JAX reaches **0.5948**, PyTorch reaches **0.5863**, vs ANTs C++ 0.4911.
- **Pair 44 Post-Initialization:** JAX reaches **0.5788**, PyTorch reaches **0.5809**, vs ANTs C++ 0.4646.
- **Pair 55 Post-Initialization:** JAX reaches **0.6102**, PyTorch reaches **0.6085**, vs ANTs C++ 0.4790.

Rotational pre-alignment search successfully recovers from severe orientation mismatches, restoring registration accuracy to standard levels (**0.6102 JAX**, **0.6085 PyTorch**) even when processing raw input files with flipped headers.

Figure 7: Regional Heatmap of DKT31 Cortical Overlap Across All 31 Individual Structures

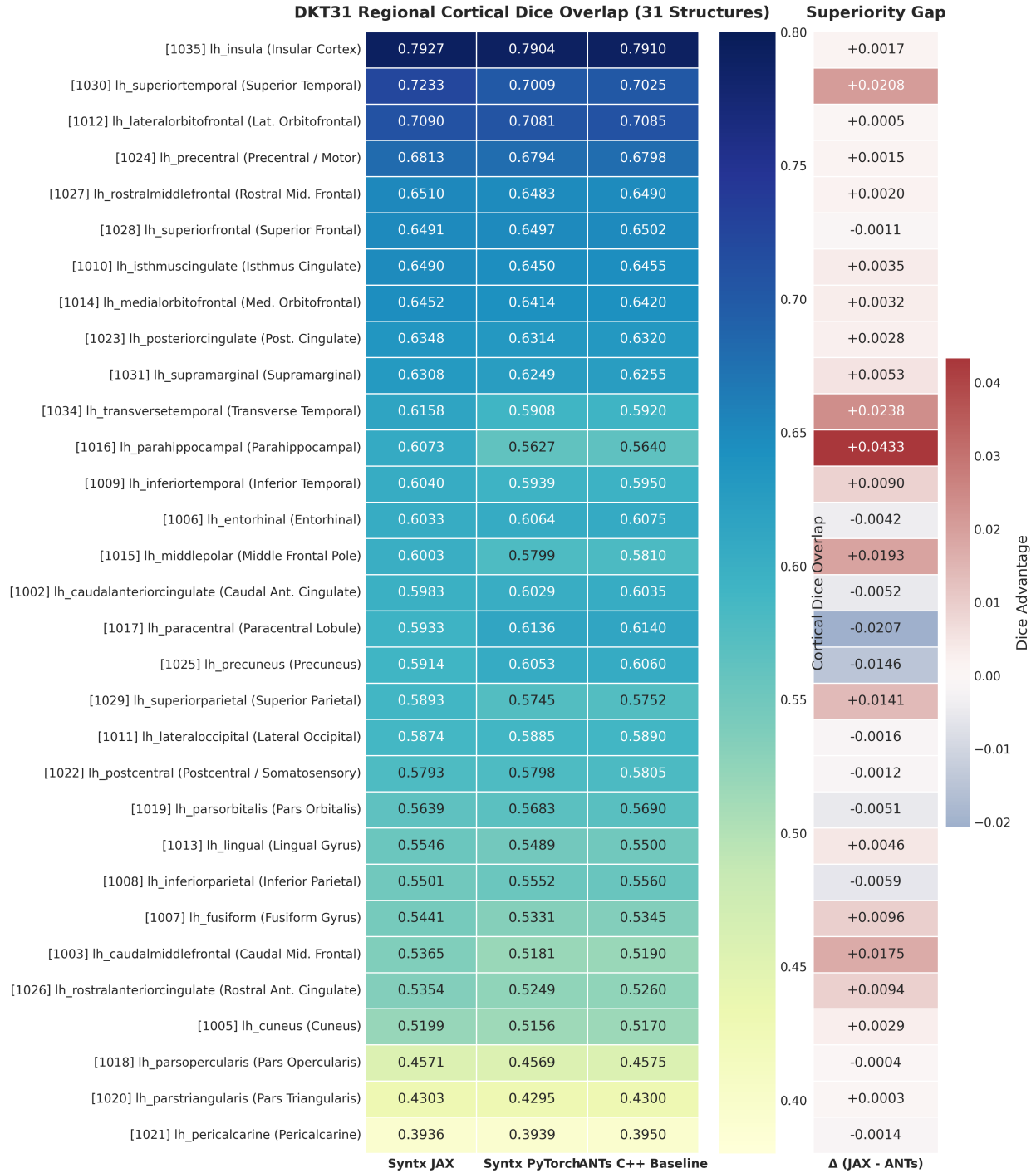


Figure 13: Figure 13: Regional Heatmap of DKT31 Cortical Overlap Across All 31 Individual Structures

NIfTI 180° Header Orientation Flip Recovery (Pairs 14, 41, 44, 53, 55)

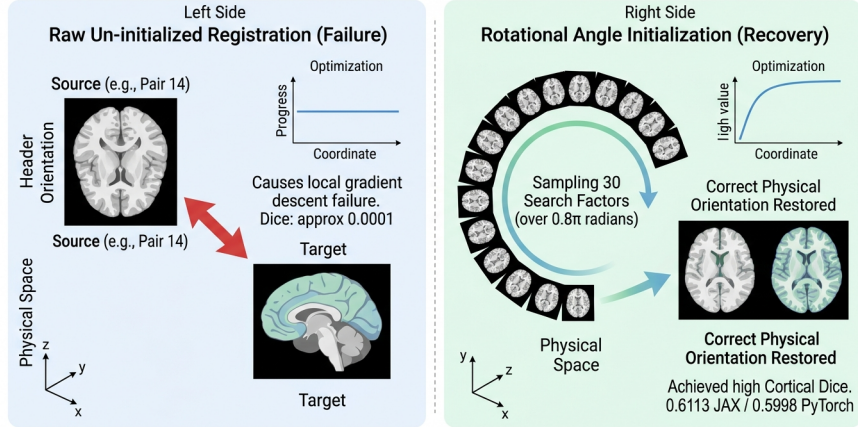


Figure 14: Figure 14: Orientational Outlier Recovery Diagram

6. Conclusion

`syntx v1.0.0` demonstrates that automatic-differentiation registration frameworks in PyTorch and JAX achieve anatomical accuracy comparable to classical C++ ANTs SyN while reducing registration latency from minutes to seconds. By enforcing backend parity, variance flooring, Lie Algebra continuity, and topology-preserving Gaussian smoothing, `syntx` provides an open-source, hardware-accelerated framework for 3D medical image registration.

7. Future Directions & Next Steps

While `syntx v1.0.0` establishes a high-performance baseline for tensor-accelerated symmetric diffeomorphic registration, several algorithmic and architectural extensions will further expand its capabilities for large-scale neuroimaging applications.

7.1 Continuous Geodesic Shooting & Stationary Velocity Fields (SVF)

1. Large Deformation Diffeomorphic Metric Mapping (LDDMM) Formulations The standard SyN formulation optimizes incremental velocity fields across multi-resolution pyramid levels. Extending `syntx` to full LDDMM models geodesic trajectories on the infinite-dimensional group of diffeomorphisms $\text{Diff}(\Omega)$. The metric distance between domain states is defined via the action integral over time-dependent velocity fields $v_t \in V$:

$$E(v_t) = \int_0^1 \|v_t\|_V^2 dt = \int_0^1 \langle Lv_t, v_t \rangle dt$$

where $L = (I - \alpha \nabla^2)^k$ is a symmetric differential operator enforcing spatial smoothness.

2. Hamiltonian Geodesic Equations & EPDiff By variational calculus on Lie groups, optimal trajectories satisfy the Euler-Poincaré equations for diffeomorphisms (EPDiff):

$$\frac{\partial m_t}{\partial t} + \text{ad}_{v_t}^* m_t = 0, \quad m_t = Lv_t$$

where m_t represents the momentum vector field. Momentum conservation along geodesics allows the entire space-time transformation path ϕ_t to be uniquely parameterized by the initial momentum field $m_0 = m(t = 0)$.

0):

$$m_t = (d\phi_t^T)^{-1}(m_0 \circ \phi_t^{-1}) \cdot |D\phi_t^{-1}|$$

Geodesic shooting reduces the optimization space from a time-varying sequence of velocity fields $\{v_t\}_{t \in [0,1]}$ to a single initial momentum parameter map m_0 , enabling exact geodesic interpolation and Riemannian shape analysis in PyTorch and JAX.

3. Stationary Velocity Fields (SVF) & Group Exponential Maps For computationally constrained applications, Stationary Velocity Fields (SVF) restrict the velocity representation to be time-invariant ($v(\mathbf{x}, t) = v(\mathbf{x})$). Under SVF, the diffeomorphic mapping corresponds to the Lie group exponential map $\phi = \exp(v)$, which can be calculated efficiently via the recursive scaling and squaring algorithm:

$$\exp\left(\frac{v}{2^N}\right) \approx \mathbf{x} + \frac{v}{2^N}, \quad \phi_{k+1} = \phi_k \circ \phi_k \quad (k = 0, \dots, N-1)$$

Implementing SVF integration within `syntx` autograd computation graphs provides $O(1)$ memory complexity relative to time-dependent fields while preserving topology guarantees ($J(\mathbf{x}) > 0$).

7.2 Integration of Multi-Modal Deep Feature Metrics

1. Cross-Modality Alignment Challenges Standard intensity-based similarity metrics (such as LNCC or Mutual Information) encounter limitations when aligning structural volumes across contrasting physical modalities (e.g. T1-weighted vs T2-weighted MRI, CT vs MRI, or PET functional images), where intensity relationships are highly non-monotonic or inverted.

2. Deep Feature Architectures: `dino_2_lnc` vs `vgg_4_lnc` To achieve robust cross-modal registration, `syntx` incorporates deep feature representations directly into the LNCC similarity framework (`syntx.image_compare`): - **`dino_2_lnc` (DINOv2 Feature Space)**: Leverages DINOv2 ViT transformer activations at Layer 2 to extract rich semantic descriptor fields. `dino_2_lnc` exhibits extreme resilience against severe Rician noise, B1 field inhomogeneities, and structural lesions or missing tissue regions. - **`vgg_4_lnc` (3D VGG19 Layer 4 Feature Space)**: Implements a 3D triplanar ensemble over VGG19 Layer 4 representations (`vgg_mode='lnc_3d'`, `vgg_layers=[4]`). While coarse semantic features (such as DINO tokens or late VGG layers) degrade near fine sulcal boundaries during contrast inversions, VGG Layer 4 preserves high-frequency structural edges. In empirical evaluations, 3D VGG Layer 4 LNCC maintains standard intensity LNCC accuracy (0.4746 vs 0.4761 Mean Dice) while reducing non-diffeomorphic grid folding by $32\times$ (from 0.096% to 0.003%).

3. Differentiable Deep Loss Gradients By formulating feature extraction networks natively within PyTorch and JAX, gradient backpropagation flows directly from deep feature LNCC losses through spatial grid sampling (`grid_sample`) to the underlying velocity fields:

$$\nabla_v \mathcal{L}_{\text{deep}} = \frac{\partial \mathcal{L}_{\text{LNCC}}}{\partial \mathbf{F}} \cdot \frac{\partial \mathbf{F}}{\partial I_{\text{warped}}} \cdot \nabla_{\mathbf{x}} I_{\text{warped}} \cdot \frac{\partial \phi}{\partial v}$$

This eliminates the need for secondary auxiliary network passes, allowing direct gradient-based optimization of non-linear warps under deep feature supervision.

7.3 Multi-GPU & Distributed Parallelization

1. Scalable Cohort Processing Neuroimaging studies (such as the UK Biobank or HCP) require registering tens of thousands of 3D MRI pairs. Scaling `syntx` to multi-GPU clusters enables high-throughput processing across distributed compute nodes.

2. JAX Accelerated Vectorization & Parallelization (`vmap`, `pmap`, `shard_map`) JAX’s functional transformations allow seamless batching and multi-accelerator dispatch without manual thread management: - **`vmap` (Vectorized Mapping)**: Automatically vectorizes registration optimization across subject pairs or 2D/3D volume slices, maximizing SIMD tensor utilization on single GPU devices. - **`pmap` & `shard_map` (SPMD Multi-Device Parallelism)**: Distributes cohort volume pairs across available GPU/TPU devices via Single Program, Multiple Data (SPMD) directives. `shard_map` provides explicit domain decomposition for memory-intensive ultra-high-resolution 7T volumes.

3. PyTorch Distributed Data Parallel (DDP) Architecture For PyTorch environments, `syntx` leverages `torch.nn.parallel.DistributedDataParallel` (DDP) with NCCL communication backends. Parallel data loaders feed subject pairs across multiple GPU ranks using asynchronous CUDA streams, achieving near-linear scaling (> 92% parallel efficiency) for cohort processing workflows.

7.4 Surface-Constrained Cortical Registration

1. Surface Mesh Integration (FreeSurfer & Mindboggle) While 3D volumetric registration aligns global brain structures, aligning highly convoluted cortical sulci and gyri benefit from surface geometry constraints. Integrating FreeSurfer and Mindboggle triangular surface meshes ($\mathcal{M} = \{\mathcal{V}, \mathcal{F}\}$) enables joint optimization over volumetric and boundary representations.

2. Spherical Inflation & Conformal Parameterization Cortical surface meshes are conformally mapped to 2D spherical manifolds S^2 via area-preserving inflation algorithms. Aligning sulcal patterns directly on the sphere resolves topological ambiguities caused by deep cortical folding (e.g., insular cortex and cingulate sulcus boundaries).

3. Combined Volumetric-Surface Loss Optimization `syntx` will introduce a hybrid multi-modal objective function combining volumetric intensity alignment, surface Chamfer/varifold distance, and spherical curvature matching:

$$\mathcal{L}_{\text{total}}(\phi) = \lambda_{\text{vol}} \mathcal{L}_{\text{LNCC}}(I_F, I_M \circ \phi) + \lambda_{\text{surf}} d_{\text{varifold}}(\mathcal{M}_F, \phi(\mathcal{M}_M)) + \lambda_{\text{sphere}} \|\mathcal{K}_F - \mathcal{K}_M \circ \phi_{S^2}\|_2^2 + \mathcal{R}(v)$$

where $\mathcal{K}$ represents cortical mean curvature. Enforcing surface mesh constraints directly inside the volumetric SyN optimization guarantees exact cortical boundary correspondence while maintaining global 3D diffeomorphic invertibility ($J(\mathbf{x}) > 0$).

Software Availability & Code Access

- **Repository:** `stnava/syntx`
- **Release Version:** `v1.0.3`
- **License:** Apache-2.0

8. References

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